03 Rolling Validation Selection Error Rate (Sliding Window)

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Goal (sliding window)

One-step-ahead **sliding** rolling validation: fixed window `w = floor(0.5 * n)`; for `t = 1, …, n - w` fit on rows `t:(t + w - 1)` and predict row `t + w`. Mean squared prediction error is averaged over `n - w` steps (same step count as expanding at the same `n` and `train_frac`). Model comparison and grid match the expanding-window analysis.

Linear truth

Simulation settings

Table 1: Rolling validation settings (sliding window, linear truth).

setting	value
seed	2026
window type	sliding
train_frac (w = floor(train_frac * n))	0.5
horizon	1
Monte Carlo replications per n	1000
Sample sizes (grid)	50, 100, …, 1000 (step 50)
beta0	1
beta1	2
sigma	4
poly_degree	3
truth	linear: y = beta0 + beta1*x + eps
selection rule	Choose model with lower mean RV MSE; tie -> linear

Error rate summary

Table 2: Monte Carlo mean selection-error rate (truth is linear; error = poly chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.004	0.002
100	0.001	0.001
150	0.000	0.000
200	0.000	0.000
250	0.000	0.000
300	0.000	0.000
350	0.001	0.001
400	0.000	0.000
450	0.000	0.000
500	0.000	0.000
550	0.000	0.000
600	0.001	0.001
650	0.001	0.001
700	0.000	0.000
750	0.000	0.000
800	0.000	0.000
850	0.000	0.000
900	0.001	0.001
950	0.000	0.000
1000	0.000	0.000

Error rate vs sample size

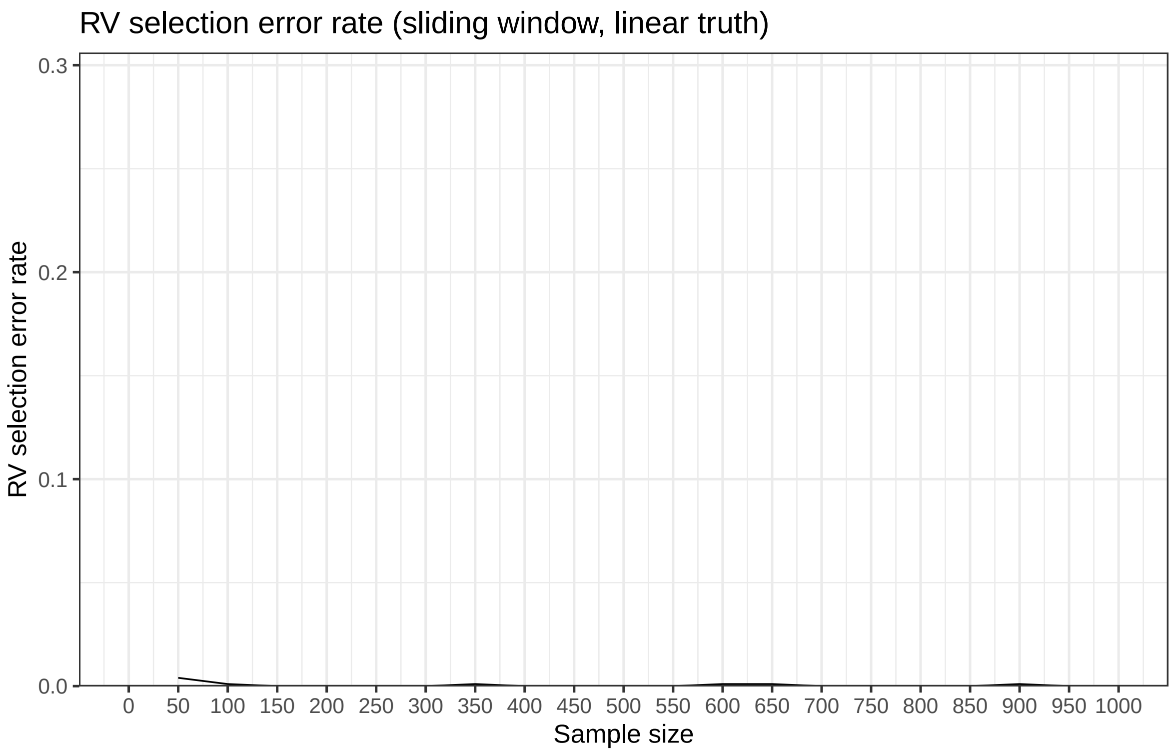

Figure 1: RV selection error rate under linear truth (sliding window).

Polynomial truth

Simulation settings

Table 3: Rolling validation settings (sliding window, polynomial truth).

setting	value
seed	2026
window type	sliding
train_frac (w = floor(train_frac * n))	0.5
horizon	1
Monte Carlo replications per n	1000
Sample sizes (grid)	50, 100, ..., 1000 (step 50)

setting	value
beta0	1
beta1	2
beta2	1
beta3	1
sigma	4
poly_degree	3
truth	polynomial: y = beta0 + beta1*x + beta2*x^2 + beta3*x^3 + eps
selection rule	Choose model with lower mean RV MSE; tie -> linear

Error rate summary

Table 4: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.940	0.00751
100	0.734	0.01397
150	0.507	0.01581
200	0.240	0.01351
250	0.124	0.01042
300	0.059	0.00745
350	0.026	0.00503
400	0.011	0.00330
450	0.003	0.00173
500	0.000	0.00000
550	0.001	0.00100
600	0.000	0.00000
650	0.000	0.00000
700	0.000	0.00000
750	0.000	0.00000
800	0.000	0.00000
850	0.000	0.00000
900	0.000	0.00000
950	0.000	0.00000
1000	0.000	0.00000

Error rate vs sample size

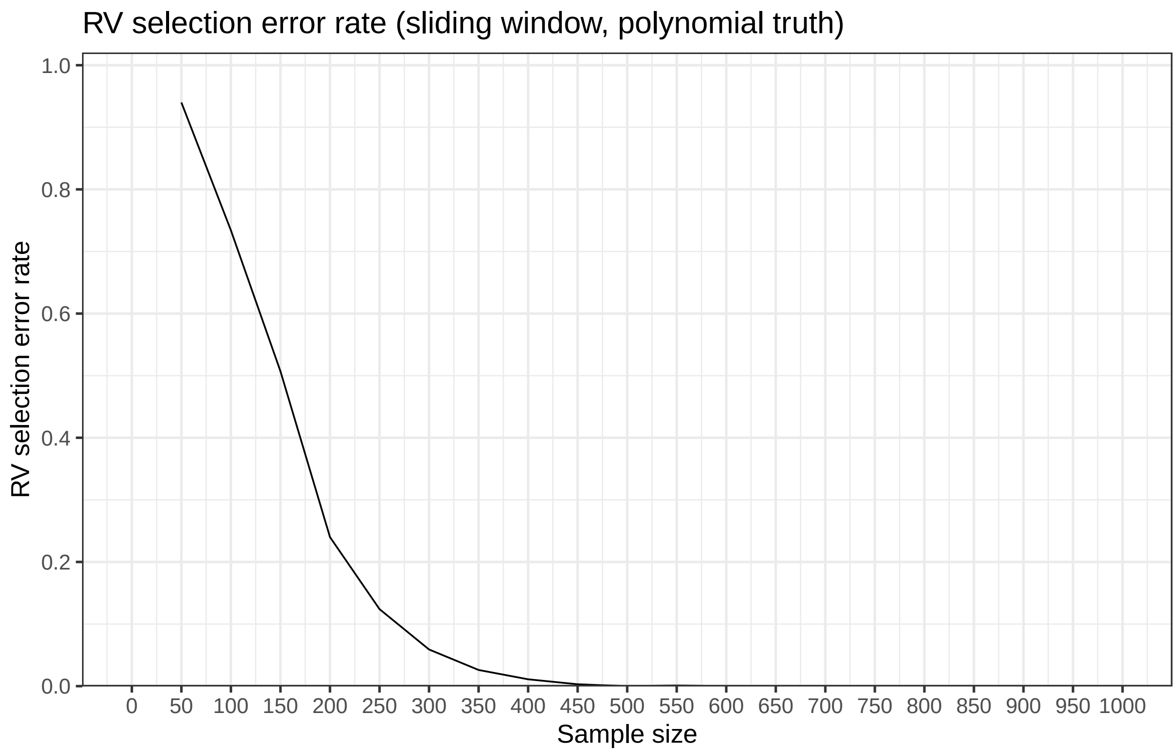

Figure 2: RV selection error rate under polynomial truth (sliding window).